RESEARCH ARTICLE

Long-Term Effect of Intensive vs Standard Blood Pressure Control on Mild Cognitive Impairment and Probable Dementia in SPRINT

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Abstract

Background and Objectives

The Systolic Blood Pressure Intervention Trial suggested that intensive lowering of systolic blood pressure (SBP) decreases the risk of developing dementia. However, an insufficient number of probable dementia cases stemming from the trial's early termination made results inconclusive. The goal of this study was to estimate the effect of intensive vs standard SBP lowering on the longer term incidence of cognitive impairment leveraging extended follow-up for cognitive status.

Methods

This is a prespecified secondary analysis of a randomized clinical trial. Between 2010 and 2013, patients aged 50 years and older with hypertension and increased cardiovascular risk excluding those with diabetes mellitus or history of stroke were recruited from 102 clinics in the United States and Puerto Rico. Participants were randomized to a SBP goal of <120 mm Hg (intensive treatment) or <140 mm Hg (standard treatment) and received treatment for 3.3 years. In-person cognitive assessment follow-up occurred through July 2018. Continued ascertainment of cognitive status by telephone began in December 2019 for participants who had not withdrawn consent or been previously adjudicated with probable dementia, but provided consent for future research. Data were analyzed using survival analyses.

Results

Of 9,361 randomized participants, 7,221 (77%) were eligible to be re-contacted. Cognitive status of 4,232 (59%) was ascertained (mean age 67 years, 36% female). We accrued a total of 216 new cases of probable dementia, less than our target of 326. Over a median follow-up of 7 years, 248 participants of the intensive treatment group (8.5 per 1,000 person-years) were adjudicated with probable dementia, compared with 293 participants (10.2 per 1,000 person-years) in the standard treatment group (hazard ratio [HR], 0.86; 95% CI, 0.72–1.02). Consistent with earlier results from the trial, the rate of both mild cognitive impairment (MCI; HR, 0.87 95% CI, 0.76–1.00) and a composite of MCI or probable dementia was lower with intensive treatment (HR, 0.89; 95% CI, 0.79, 0.99).

Discussion

Among ambulatory adults with hypertension and high cardiovascular risk, intensive treatment vs standard treatment of SBP for 3.3 years resulted in a lower risk of MCI and cognitive impairment including MCI or probable dementia, but not for probable dementia alone.

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Glossary

HR = hazard ratio; **MCI** = mild cognitive impairment; **SBP** = systolic blood pressure; **SPRINT** = Systolic Blood Pressure Intervention Trial.

Classification of Evidence

This study provides Class II evidence that intensively reducing SBP (target <120 mm Hg) decreases the risk of cognitive impairment in individuals aged 50 years and older with hypertension.

Clinical Trial Information

Clinical trial number NCT01206062.

Introduction

Dementia is a syndrome affecting memory and other cognitive functions, behavioral functioning, and social abilities that impair daily life. Previous studies estimate that over 9 million Americans could have dementia by 2030 and nearly 12 million by 2040.[1] The prevalence of mild cognitive impairment (MCI), a transitional state between normal cognitive aging and dementia, is also expected to increase.[2]

Hypertension affects more than 50% of persons in the United States by the age of 50 years and 75% by age 65 or older[3] and has also been identified as a modifiable risk factor for the prevention of dementia.[4-6] Cerebrovascular disease underlies vascular cognitive impairment and is commonly present in Alzheimer dementia, where it frequently co-occurs with β-amyloid and tau neuropathology.[7,8]

Randomized clinical trials of hypertension management have shown variable effects on cognitive function and the incidence of dementia, although those trials have tended to have follow-up less than 4 years and have had wide variability in achieved blood pressure levels because lower blood pressure targets have been pursued in the past 30–40 years.[9,10] Results from the Systolic Blood Pressure Intervention Trial (SPRINT) demonstrated a beneficial effect of intensive systolic blood pressure (SBP) lowering on mild cognitive impairment (MCI) but were inconclusive for probable dementia.[11] Owing to the increasing prevalence of hypertension in middle-aged and older adults and the growing personal and societal impact of dementia,[3] there is a clear need for robust long-term results from randomized trials of SBP-lowering therapy to inform health care policy and promote greater adherence to guideline-based strategies for hypertension management.[12]

The goal of this study was to address a limitation of the SPRINT results regarding probable dementia, insufficient follow-up to accrue enough incident cases of dementia to provide a well-powered test comparison of intensive vs standard BP lowering, and to assess the long-term impact on cognitive function of only 3.3 years of intensive blood pressure control.

The primary research question for this study was whether intensively reducing SBP would decrease the longer term incidence of cognitive impairment in the SPRINT cohort.

Methods

Main Trial Design and Participants

The trial design and methods for the main trial and results for the primary cardiovascular outcomes and cognitive outcomes, which were prespecified secondary outcomes, have been published previously.[11,13,14] SPRINT was a randomized clinical trial comparing a SBP target of <120 mm Hg (intensive treatment) with a target of <140 mm Hg (standard treatment). Between 2010 and 2013, patients aged 50 years and older with hypertension and increased cardiovascular risk excluding those with diabetes mellitus, history of stroke, or polycystic kidney disease were recruited from 102 clinic sites in the United States and Puerto Rico. The trial's intervention was terminated early, when participants had received treatment for an average of 3.3 years, because of a beneficial effect of intensive treatment on its primary outcome, incident cardiovascular disease and all-cause mortality. In-person cognitive assessment follow-up occurred through July 2018.

This study was ancillary, continuing ascertainment of cognitive status beginning in December 2019 for participants in the main trial cohort who had not withdrawn consent, had not been previously adjudicated with probable dementia, and had provided informed consent for future research (Figure 1). Using a centralized validated telephone-based approach,[15] we performed additional follow-up for cognitive status, increasing the median follow-up in the cohort by approximately 2 years to approximately 7 years. The study was designed to ascertain at least 326 new cases of probable dementia.

Standard Protocol Approvals, Registrations, and Patient Consents

The study was approved by the Wake Forest University School of Medicine Institutional Review Board, and all participants who were contacted by telephone were asked to provide verbal informed consent. The Systolic Blood Pressure Intervention

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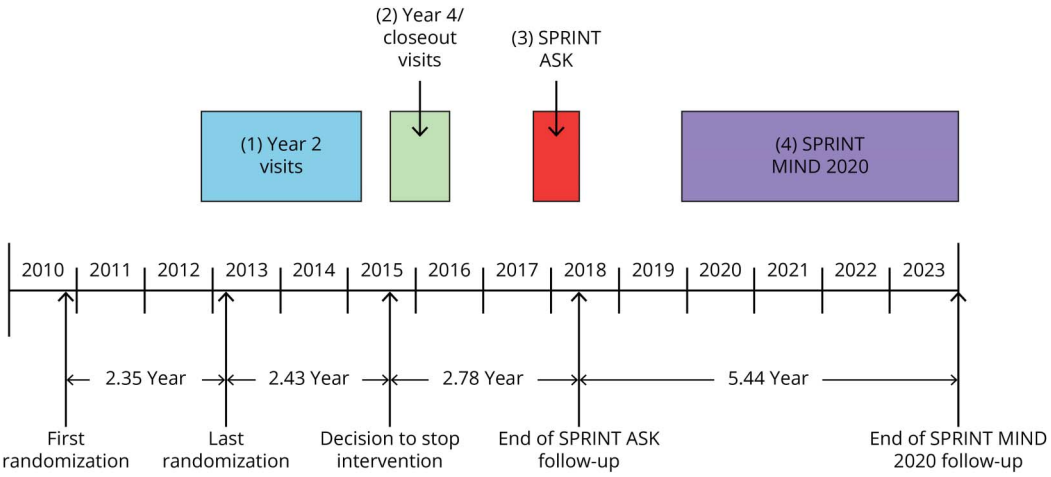
Figure 1 Time Line for Follow-Up in the Systolic Blood Pressure Intervention Trial (SPRINT)



Participants in the trial could have up to 4 cognitive assessments during the course of the trial and observational follow-up. SPRINT ASK denotes the Alzheimer's, Senior, and Kidney follow-up study.

Trial was approved by institutional review boards at each participating site, and each participant provided written informed consent. This study is registered on ClinicalTrials.gov as NCT01206062.

Ascertainment of Cognitive Status

Procedures for neuropsychological evaluation and adjudication of cognitive impairment in the trial have been previously described.[11] The entire period of follow-up used expert adjudication to classify participants as either no impairment, MCI, probable dementia, or unable to classify. For the on-treatment phase of the trial, adjudicators relied on data from clinic-based in-person neuropsychological assessments, interviews with a trusted contact (family member or friend familiar with their day-to-day functioning), medical information including comorbidities and medications, and assessment of depression. Assessments for participants no longer able to complete face-to-face visits were performed by telephone using a standard protocol or with their trusted contacts. After the conclusion of the treatment phase, the long-term follow-up data were collected using centralized telephone follow-up assessments following standardized and validated procedures used in other studies with the same outcomes.[15,16] This form of data collection also allowed for continued ascertainment throughout the COVID-19 pandemic. Standardized assessments were administered to participants by trained and certified examiners and included the Modified Telephone Interview for Cognitive Status,[17] the Hopkins Verbal Learning Test–Revised,[18] Logical Memory test,[19] Digit Span test,[20] Category Fluency-Animals,[21] Oral Trail Making Test,[22] and the 8-item Patient Health Questionnaire,[23] a measure of depressive symptoms. The Functional Activities Questionnaire,[24] a measure of functional status related to cognitive abilities, was administered to each participant's selected trusted contact by telephone. If a participant had died or was otherwise unable to complete the phone-based cognitive assessment, the Dementia Questionnaire,[25] a validated, structured interview assessing observed cognitive and behavioral changes, functional status, and medical and health information, was administered to the contact. For all tests and questionnaires, validated Spanish translations were used when available. Otherwise, instruments were translated by native Spanish speakers and then back translated.

Adjudication of cognitive status was completed by a panel of experts that included those involved in the trial period and consisted of neurologists, geriatricians, geropsychiatrists, neuropsychologists, and geropsychologists with clinical expertise diagnosing cognitive impairment. Adjudication procedures were the same throughout all phases of the trial and follow-up and used all available data collected during each phase, which, in addition to the cognitive assessments and proxy reports, included standardized measures of self-reported perceived health status, quality of life,[26] current medications, medical problems, and current health habits (smoking, alcohol use, and physical activity). All references to the treatment group were redacted, and the adjudicators were masked to treatment assignment. Participants scoring above 31 on the TICS-M were presumed to have no cognitive impairment; those with scores ≤31 were referred for adjudication.[27] Participants were then adjudicated as having either no cognitive impairment, MCI, or probable dementia. Unclassifiable cases (e.g., significant depression and recent health event) were placed in a "cannot classify" category. Cases were reviewed by 2 adjudicators independently using standardized diagnostic criteria for probable dementia and MCI.[28,29] When the 2 adjudicators agreed, their decisions were considered final. Disagreements were discussed by the full panel, and the classification decision was made by a majority vote of the panel members.

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Duration of Follow-Up

The main trial included planned cognitive assessments at baseline, 2 and 4 years of follow-up, and study closeout if it was more than 1 year after the 4-year follow-up visit (Figure 1). The Alzheimer's, Senior and Kidney follow-up visit was conducted between October 2017 and July 2018 and included the same in-person neuropsychological assessment used during the main trial, or, for those unable to attend the clinic visit, a telephone-based cognitive assessment. This study conducted additional telephone assessments beginning in December 2019.

Statistical Analysis

Similar to the main trial, the primary cognitive outcome for this study was occurrence of adjudicated all-cause probable dementia. Secondary cognitive outcomes included occurrence of adjudicated MCI and a composite outcome of occurrence of either probable dementia or MCI. The occurrence of MCI was defined as 2 or more consecutive adjudicated classifications of MCI. The analysis included all available adjudicated cognitive outcomes from the cohort during both the main trial and this extended follow-up study. The primary hypothesis was that incidence of probable dementia over the entire follow-up period would be lower for participants assigned to intensive treatment compared with those assigned to standard treatment. We expected to conduct study assessments in at least half of the eligible cohort, which we estimated would ascertain 326 new cases of probable dementia (651 cases total) and provide 90% power for detecting a 17% relative reduction (hazard ratio [HR] of 0.83).

The cumulative incidence of MCI, probable dementia, and the composite of both outcomes was estimated separately accounting for the competing risk of death.[30] Treatment group comparisons were conducted using cause-specific Cox proportional hazard regression models with the baseline hazard function stratified by clinic.[31,32] Interactions between the treatment group and subgroups of interest were assessed with a likelihood ratio test. Subgroups included age (<75 vs ≥75 years), sex, race (Black vs non-Black), education (<high school education, high school education or general education diploma, or > high school education), and baseline global cognitive function based on the Montreal Cognitive Assessment.[33] We classified normal vs lower cognitive function using age-specific and education-specific thresholds modeled after normative data from the Irish Longitudinal Study of Ageing[34,35] (eTable 1).

Given previous evidence that the benefit of intensive treatment on all-cause mortality observed during the trial attenuated over the course of extended follow-up,[36] we conducted sensitivity analyses estimating a time-dependent coefficient for intensive treatment within a cause-specific Cox regression framework.[37] We also used this framework to examine whether there was differential ascertainment of cognitive status between the treatment groups by estimating whether the censoring distribution differed over time between the treatment groups. All hypothesis tests were 2-sided and performed at the $\alpha = 0.05$ level of significance. There was no adjustment of the significance threshold for the secondary or other end points. All analyses were performed using SAS version 9.4 (SAS Institute Inc) and the R Statistical Computing Environment.[38]

Data Availability

The primary data from SPRINT are available in the National Heart, Lung, and Blood Institute BioLINCC Repository.[39] The additional follow-up data reported here for cognitive status will be available by December 31, 2025.

Results

Study Participants

Characteristics of the 9,361 randomized participants in the main trial have been published previously.[11,14] In brief, participants had a mean age of 68 years (SD, 9.4 years), and 36% were female, 30% Black, and 11% Hispanic. Most of the participants had high educational attainment, 90% with at least a high school education and 39% with a college degree or higher, and the median Montreal Cognitive Assessment score at baseline was 23 (interquartile range, 20–26).

Follow-Up Ascertainment of Cognitive Status

Of the 9,361 randomized participants in the main trial, 97% had at least one follow-up cognitive assessment and 7,221 (77.1%) were eligible to be contacted for extended follow-up to ascertain their cognitive status (Figure 2). The eligible cohort was similar to the full trial cohort (Table 1). Phone calls began on December 2, 2019, and ended on December 31, 2023. Of the 7,221 participants in the cohort who were eligible to be contacted, cognitive status was ascertained during this extended follow-up in 4,232 participants (58.6%) at least once. A small number of participants (N = 149, 3.5%) were ascertained twice as part of this extended follow-up. Completion rates for the ascertainment of cognitive status were not statistically different between the treatment groups (Table 2). In general, participants who did not complete a cognitive assessment during this extended period of follow-up were more likely to be Black and had higher baseline SBP, lower estimated glomerular filtration rate, and less use of aspirin at baseline (Table 2). The occurrence of indeterminate adjudications (i.e., decisions of "cannot classify") was low (N = 173, 12%) and was also similar between the treatment groups ($p = 0.38$). At the end of follow-up, 69% of the cohort had been either censored or observed to have no event and 21% had died before being adjudicated to have MCI or probable dementia (eFigure 1).

Primary Cognitive Outcome

A total of 216 new adjudications of probable dementia were ascertained during this phase of extended follow-up, increasing the total number of probable dementia cases to 541. This was less than the target of 651 in our original design. Over a median follow-up of 6.9 years (interquartile range 4.0–9.6 years), a total of 248 participants randomized to intensive treatment were adjudicated with probable dementia (8.5 per 1,000 person-years), compared with 293 participants (10.2 per 1,000 person-years) in the standard treatment group

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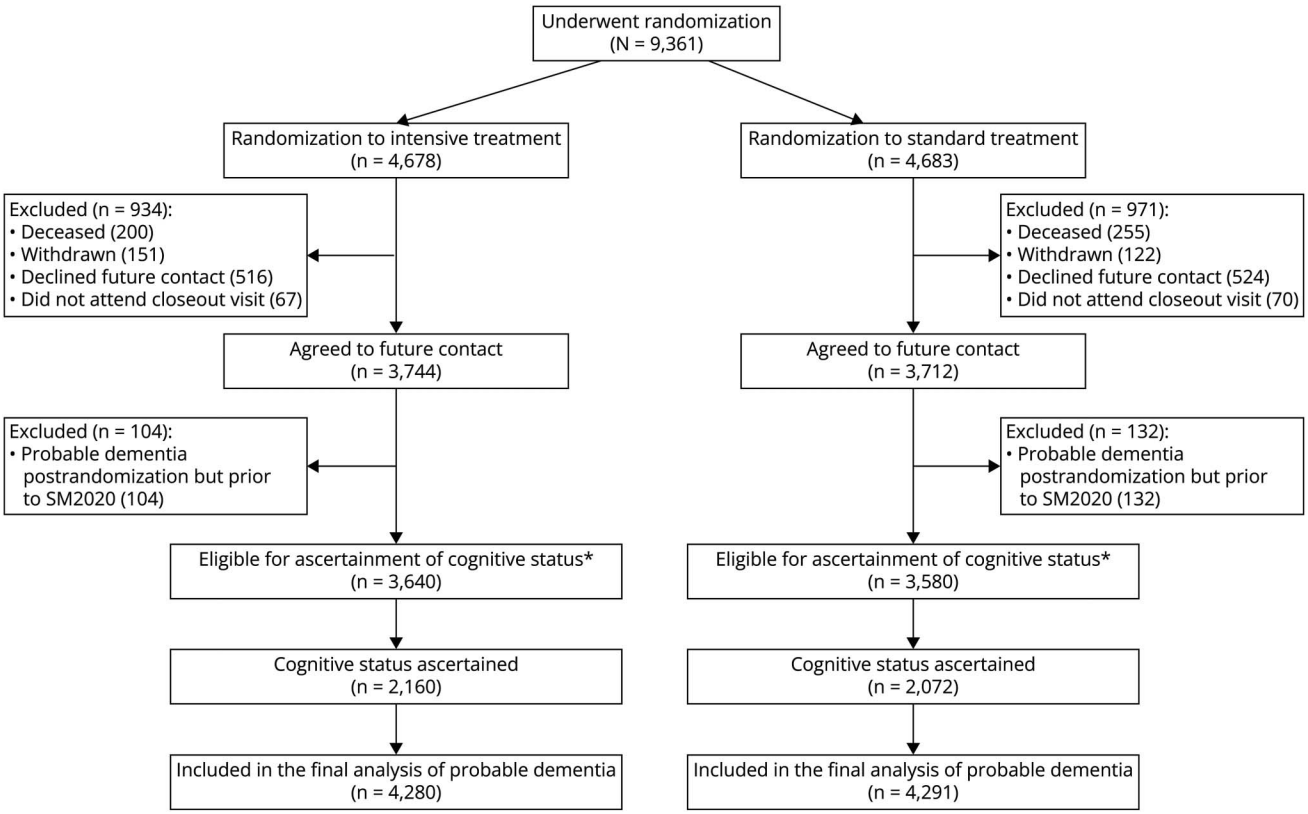
Figure 2 Randomization and Follow-Up for Cognitive Outcomes in the Systolic Blood Pressure Intervention Trial



(HR, 0.86; 95% CI, 0.72–1.02) (Table 3 and Figure 3). When we examined the proportional hazard assumption using Schoenfeld residuals, they indicated an increasing difference between the treatment groups during the later phase of follow-up, although the proportional hazard assumption did not seem to be severely violated ($p = 0.80$) (Figure 3). The incidence of mortality exceeded the incidence of probable dementia and the composite outcome of MCI or probable dementia later in the follow-up period (eFigure 2). When death was treated as a competing risk using a Fine-Gray subdistribution hazard model, the results regarding the effect of intensive treatment on probable dementia were highly similar (HR, 0.86; 95% CI, 0.73–1.01). There was no evidence of treatment effect heterogeneity across the subgroups of interest (eFigure 3). As expected, participants aged 75 years or older at baseline accounted for most of the cases of probable dementia regardless of the treatment group; however, the hazard ratio estimate in participants younger than 75 years indicated a highly similar relative reduction compared with older trial participants (eFigure 3).

Secondary Cognitive Outcomes

Two consecutive adjudications of MCI occurred in 380 participants in the intensive treatment group and 430 participants in the standard treatment group (HR, 0.87; 95% CI, 0.76–1.00, Table 3). There was a significant difference in the composite outcome of MCI or probable dementia favoring the intensive treatment group (20.1 vs 22.9 per 1,000 person-years; HR, 0.89; 95% CI, 0.79–0.99). As with probable dementia, there was no evidence of treatment effect heterogeneity according to subgroups of interest for either MCI alone or the composite outcome of MCI or probable dementia (eFigures 4 and 5).

Classification of Evidence

This study provides Class II evidence that intensively reducing SBP (target <120 mm Hg) decreases the risk of cognitive impairment in individuals aged 50 years and older with hypertension.

Discussion

This extended observational follow-up in SPRINT was designed to ascertain the effect of 3.3 years of intensive blood pressure control on longer term cognitive status in the cohort, with the specific goal of increasing the number of ascertained cases of probable dementia enough to provide a robustly powered test of the effect of intensive SBP lowering. Additional objectives included evaluating the legacy effect of intensive treatment on the reductions observed in the trial's primary report for MCI and the composite outcome of MCI or probable dementia. Over a median of almost 7 years of follow-up, we observed that the previously reported statistically significant reduction in the rate of cognitive impairment (composite of MCI or probable dementia) was maintained. The estimated effect on probable dementia, although not statistically significant, was also similar to the primary trial

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Table 1 Baseline Characteristics of SPRINT Clinical Trial Participants and Participants Eligible for Additional Ascertainment

	SPRINT N = 9,361	Eligible for additional ascertainment N = 7,220	*p* Value
Randomized to intensive treatment, no. (%)	4,678 (50.0)	3,640 (50.4)	0.57
Age, mean (SD), y	67.93 (9.44)	67.4 (9.0)	0.0001
Age 75 y or older, no. (%)	2,636 (28.2)	1817 (25.2)	<0.001
Female sex, no. (%)	3,332 (35.6)	2,515 (34.8)	0.31
Race/ethnicity, no. (%)			0.51
Black	2,802 (29.9)	2,126 (29.4)	
Hispanic	984 (10.5)	722 (10.0)	
Other	176 (1.9)	130 (1.8)	
White	5,399 (57.7)	4,242 (58.8)	
Seated blood pressure, mean (SD), mm HG			
Systolic	139.7 (15.6)	139.3 (15.3)	0.13
Diastolic	78.1 (11.9)	78.4 (11.8)	0.09
Orthostatic hypotension, no. (%)	685 (7.3)	505 (7.0)	0.42
History of cardiovascular disease, no. (%)	1877 (20.1)	1,386 (19.2)	0.17
Estimated GFR, mean (SD), mL/min/1.73 m^2	71.75 (20.60)	72.6 (20.2)	0.01
Total cholesterol, mean (SD), mg/dL	190.11 (41.17)	190.3 (41.4)	072
HDL cholesterol, mean (SD), mg/dL	52.87 (14.47)	52.6 (14.2)	0.29
Glucose, mean (SD), mg/dL	98.81 (13.55)	98.9 (13.3)	0.54
Statin use, no. (%)	4,054 (43.7)	3,120 (43.5)	0.81
Aspirin use, no. (%)	4,756 (51.0)	3,687 (51.2)	0.83
BMI, mean (SD), mg/dL	29.85 (5.77)	30.0 (5.8)	0.03

analysis in showing a lower incidence of probable dementia associated with intensive treatment. While these dementia results are not conclusive, it is important to remember that participants randomized to intensive treatment underwent an average of just 3.3 years of intervention, after which the same level of intensive treatment of hypertension was generally not maintained.[36] Thus, these data support intensive BP treatment as an important strategy for the prevention of cognitive impairment and suggest some degree of persistent benefit on the development of cognitive impairment from even a few years of intensive treatment.

While SPRINT and several previous trials have shown that better management of hypertension consistently benefits cardiovascular morbidity and mortality,[11,40] meta-analyses of randomized trials have in totality supported a beneficial effect of BP control on dementia risk.[41-43] The conclusions of these meta-analyses are further supported by the cognitive results from the China Rural Hypertension Control Project reported at the 2023 American Heart Association Scientific Sessions, which investigated a village doctor–led intensive BP control program vs usual care in 326 villages in rural China.[40] Over 4 years of follow-up, SBP was 20 mm Hg lower in the intensive treatment group compared with usual care, with the intensive treatment group also having a statistically significant 15% lower prevalence of dementia. Nevertheless, some caution should be exercised in interpreting these updated results from SPRINT, given that a robust reduction in probable dementia was not observed. While MCI is a strong risk factor of dementia, it exhibits variable progression to dementia with up to 40% of adults exhibiting performance consistent with normal cognitive function on subsequent testing although these persons still possess a greater probability for future incident dementia relative to persons never experiencing MCI.[2] Nevertheless, for this reason, 2 consecutive MCI adjudications were required to classify MCI in SPRINT.

The persistence of the protective effect of intensive treatment on the composite outcome of MCI or probable dementia stands in contrast to what has been observed with longer term follow-up for other outcomes in SPRINT. The blood pressure difference between the groups rapidly deteriorated to the

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Table 2 Baseline Characteristics of Participants Eligible for Ascertainment of Cognitive Status Overall and Stratified by Completion of the Cognitive Assessment

	Eligible for contact N = 7,220	Cognitive status ascertained N = 4,232	Cognitive status not ascertained N = 2,988	*p* Value
Randomized to intensive treatment, no. (%)	3,640 (50.4)	2,160 (51.0)	1,480 (49.5)	0.21
Age, mean (SD), y	67.4 (9.0)	67.4 (8.8)	67.3 (9.3)	0.53
Age 75 y or older, no. (%)	1817 (25.2)	1,037 (24.5)	780 (26.1)	0.12
Female sex, no. (%)	2,515 (34.8)	1,530 (36.2)	985 (33.0)	0.005
Race/ethnicity, no. (%)				<0.001
Black	2,126 (29.4)	1,149 (27.2)	977 (32.7)	
Hispanic	722 (10.0)	351 (8.3)	371 (12.4)	
Other	130 (1.8)	74 (1.7)	56 (1.9)	
White	4,242 (58.8)	2,658 (62.8)	1,584 (53.0)	
Seated blood pressure, mean (SD), mm HG				
Systolic	139.3 (15.3)	138.8 (15.1)	140.0 (15.6)	0.001
Diastolic	78.4 (11.8)	78.3 (11.6)	78.6 (11.9)	0.29
Orthostatic hypotension, no. (%)	505 (7.0)	297 (7.0)	208 (7.0)	0.93
History of cardiovascular disease, no. (%)	1,386 (19.2)	783 (18.5)	603 (20.2)	0.07
Estimated GFR, mean (SD), mL/min/1.73 m^2	72.6 (20.2)	72.7 (19.3)	72.4 (21.5)	0.47
Total cholesterol, mean (SD), mg/dL	190.3 (41.4)	189.8 (40.4)	191.1 (42.8)	0.17
HDL cholesterol, mean (SD), mg/dL	52.6 (14.2)	52.7 (13.9)	52.6 (14.7)	0.72
Glucose, mean (SD), mg/dL	98.9 (13.3)	99.0 (13.2)	98.9 (13.4)	0.98
Statin use, no. (%)	3,120 (43.5)	1831 (43.5)	1,289 (43.4)	0.90
Aspirin use, no. (%)	3,687 (51.2)	2,228 (52.8)	1,459 (48.9)	0.001
BMI, mean (SD), mg/dL	30.0 (5.8)	30.0 (5.8)	30.1 (5.7)	0.43

Abbreviations: BMI = body mass index; GFR = glomerular filtration rate; HDL = high-density lipoprotein.

higher BP goal after the trial ended, leading to the protective effect of intensive treatment on cardiovascular and all-cause mortality quickly attenuating after the conclusion of the trial.[36] It is encouraging to observe persistent benefit on cognitive function even when based on only 3.3 years of active treatment, also in comparison with the group with reasonably well-managed BP to a target <140 mm Hg (achieved mean SBP of 135 mm Hg). Even with newer guidelines recommending blood pressure treatment to <130 mm Hg,[12] less than 50% of patients with hypertension are controlled to a level of <140 mm Hg.[44] This raises the question of what the magnitude of public health benefit on cognitive impairment would be relative to the current suboptimal control of hypertension prevalent in most communities.

Combined with cardiovascular benefits observed during the trial and the cumulative evidence from other trials, these data lend additional support to pursuing more intensive BP control in older adults, perhaps to levels lower than most current guidelines.[12] It is important to note that the SPRINT intervention was safe; it did not show adverse impacts on several key functional and physiologic outcomes relevant to older adults including declines in gait speed, injurious falls, and reduced cerebral brain perfusion.[6,45-47] In addition, several secondary analyses using SPRINT data have provided further insights that raise hypotheses for future research. Greater cognitive benefit occurred in participants who had higher baseline risk of cognitive decline,[48] and more time in the SBP target range of 110–130 mm Hg in the first 3 months after randomization was associated with lower rates of cognitive impairment.[49]

This study has several limitations. First, the BP intervention was terminated early because of its benefit on cardiovascular event and mortality incidence, resulting in a reduction of the SBP difference between the treatment groups and a loss of power to detect the effect of BP lowering on the longer term incidence of dementia. In addition, we did not reach our target

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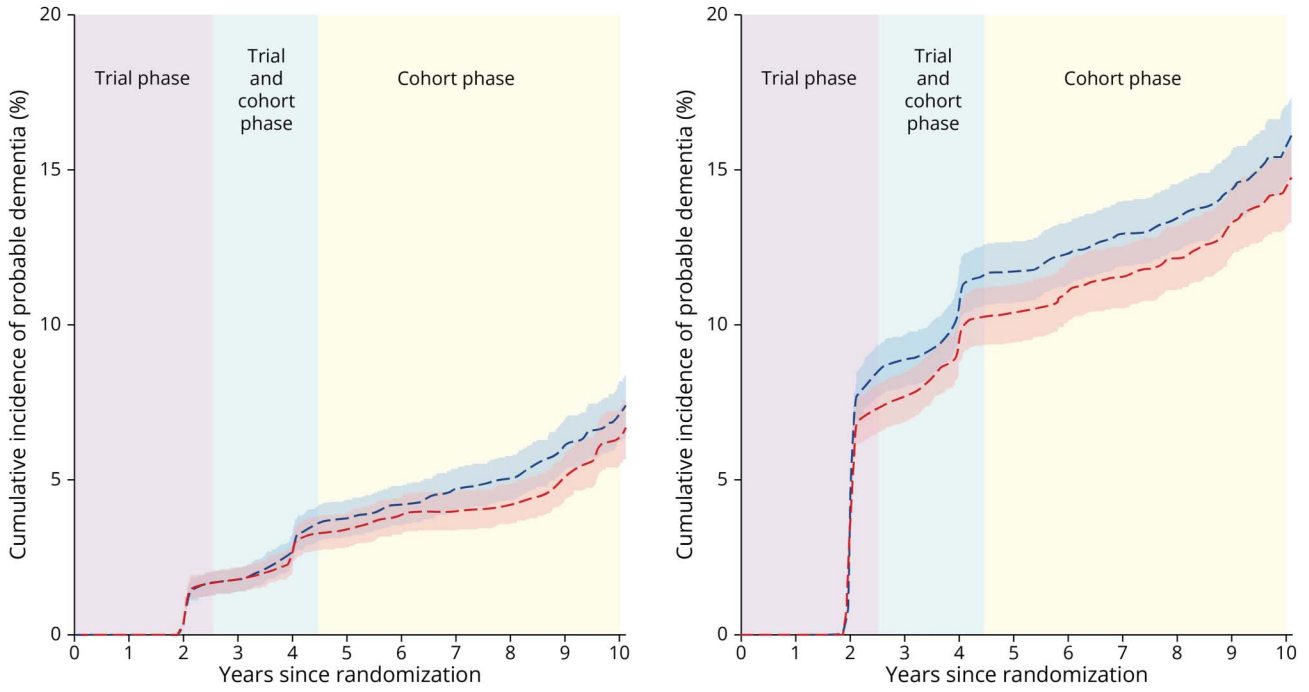
Table 3 Incidence of Probable Dementia, Mild Cognitive Impairment, and a Composite of Both Outcomes by Treatment Group

Outcomes		Intensive treatment		Standard treatment			
		No. with outcome/ person-years	Cases per 1,000 person-years	No. with outcome/ person-years	Cases per 1,000 person-years	Hazard ratio (95% CI)[b]	*p* Value
Probable dementia	Original report[a]	149/20,569	7.2	176/20,378	8.6	0.83 (0.67–1.04)	0.10
	With additional follow-up	248/29,128	8.5	293/28,866	10.2	0.86 (0.72–1.02)	0.08
Mild cognitive impairment	Original report[a]	287/19,690	14.6	353/19,281	18.3	0.81 (0.69–0.95)	0.007
	With additional follow-up	380/27,155	14.0	430/26,526	16.2	0.87 (0.76–1.00)	0.04
Mild cognitive impairment or probable dementia	Original report[a]	402/19,873	20.2	469/19,488	24.1	0.85 (0.74–0.97)	0.01
	With additional follow-up	555/27,628	20.1	622/27,122	22.9	0.89 (0.79–0.99)	0.04

[a] Results as reported in The SPRINT MIND Investigators for the SPRINT Research Group. *JAMA*. 2019; 321(6):553-61.
[b] Intensive treatment group vs standard treatment group based on a cause-specific Cox proportional hazard regression model.

of 641 cases of probable dementia for analysis. Second, the trial did not enroll persons with type 2 diabetes, prevalent stroke, advanced kidney disease, or symptomatic heart failure, limiting generalization of results to these populations. Third, the specific choice of thresholds for the Montreal Cognitive Assessment and the Modified Telephone Interview for Cognitive

Figure 3 Incidence of Probable Dementia and a Composite Outcome of Mild Cognitive Impairment or Probable Dementia in the Systolic Blood Pressure Intervention Trial (SPRINT)



Blue line indicates the standard arm, and red line indicates the intensive arm. The trial phase and cohort phase do not overlap for individual participants, but owing to participants being randomized over time, there is an overlap in the trial and cohort phase for the trial population when time is measured relative to the date of randomization. The trial phase encompasses follow-up through the end of study closeout visits (Figure 1).

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Status to trigger additional testing and adjudication may have underestimated the frequency of MCI; however, there is no indication that such an under-ascertainment differed by the treatment group. Fourth, loss to follow-up, although similar for both treatment groups, undoubtedly led to an under-ascertainment of cognitive outcomes. Fifth, while prevalent dementia was an exclusion criterion for enrollment, the trial did not adjudicate baseline cognitive status; therefore, we cannot exclude or examine the possible influence of prevalent MCI at the time of randomization, although this is doubtful given the large sample size and the many similar baseline characteristics. Sixth, we used validated telephone-based cognitive assessment procedures, rather than in-person neuropsychological assessments, and when participants could not be reached or assessed, we relied on proxy respondents. This was necessary to reach participants during this follow-up but may have introduced a bias, although similar validated procedures have been used in other studies,[15,16,50,51] and any impact would have been similar in both treatment groups.

Among ambulatory adults with hypertension and with high cardiovascular risk, treating to an SBP goal of less than 120 mm Hg compared with a goal of less than 140 mm Hg for 3.3 years results in long-term risk reduction for MCI and MCI or probable dementia, but not for probable dementia alone after 7 years of follow-up. Because of early termination of the trial's intervention, the study may have been unable to discern the full impact of improved BP control on preventing cognitive decline and dementia.

Author Contributions

D.M. Reboussin: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S.A. Gaussoin: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. N.M. Pajewski: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B.C. Jaeger: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B. Sachs: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S.R. Rapp: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M.A. Supiano: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M.L. Cleveland: major role in the acquisition of data; study concept or design; analysis or interpretation of data. V. Hunter: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. J.L. Demons: major role in the acquisition of data; study concept or design. P.K. Ogrocki: major role in the acquisition of data; study concept or design. A.J. Lerner: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. G.J. Chelune: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. V.G. Wadley: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M.L. Scales: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. N.F. Woolard: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. L.H. Perdue: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. K.E. Callahan: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.D. Williamson: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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